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Chapter 1

Mean Polytopes and Maximum Entropy Distributions

Abstract Beyond elementary: Study generic maximum entropy distributions. Motivation from the chapter before

1.1 Polytopes of Mean Parameters

We in this section investigate properties of probability distributions based on their mean parameters. Given a statistic t , we first define a mean parameter to any distribution by the expectation of the statistic.

Definition 1.1 Let there be a statistic t and a boolean base measure $\nu [X_{[d]}]$. We call the tensor

$$\mu [L] = \langle \mathbb{P} [X_{[d]}], \sigma^t [X_{[d]}, L] \rangle_{[L]}$$

the mean parameter tensor to a distribution $\mathbb{P} [X_{[d]}]$. The set

$$\mathcal{M}_{t,\nu} = \left\{ \langle \mathbb{P}, \sigma^t, \nu \rangle_{[L]} : \mathbb{P} [X_{[d]}] \in \Lambda^{\delta, \text{MAX}, \nu} \right\}$$

is called the polytope of realizable mean parameters. Here we denote by $\Lambda^{\delta, \text{MAX}, \nu}$ the set of all probability distributions representable with respect to ν (see Def. ??).

While introduced here as a property of a distribution, the mean parameters will be central to probabilistic inference in Chapter ?? We in the reminder of this section prepare for this application and derive tensor network representations for distributions having sufficient statistics t , depending on their corresponding mean parameter in the polytope.

1.1.1 Representation by Convex Hulls

First of all, we provide a simple characterization of the sets of mean parameters as the convex hull of the slices to the selection encoding of the statistic (see Figure 1.1). Convex hulls of finite vectors are called $\mathcal{V}$ -polytopes (see Lecture 1 in [?]).

Theorem 1.1 *For any statistic t the polytope of mean parameters is the convex hull of the slices of σ^t with fixed indices to $X_{[d]}$, that is*

$$\mathcal{M}_{t,\nu} = \text{conv} \left(\sigma^t [X_{[d]} = x_{[d]}, L] : x_{[d]} \in \bigtimes_{k \in [d]} [m_k], \nu [X_{[d]} = x_{[d]}] = 1 \right).$$

Proof. First we realize that the characterization of by ν representable distributions is a standard simplex extended by trivial coordinates, that is

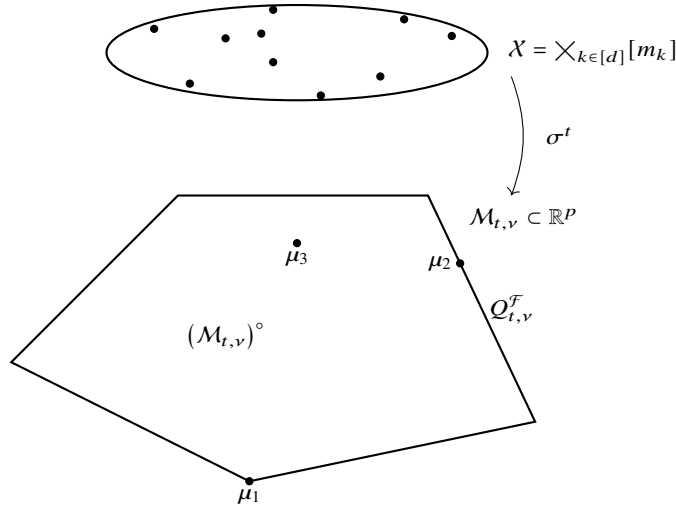


Fig. 1.1 Sketch of the mean polytope $\mathcal{M}_{t,v}$ to a statistic t , which is minimal with respect to ν . The mean polytope is a bounded subset of $\mathbb{R}^P$ (here sketched as a 2-dimensional projection). For any vertex μ_1 we find $x_{[d]}$ such that $\mu_1[L] = \sigma^t[X_{[d]} = x_{[d]}, L]$. Generic mean parameters μ_2 outside the interior are on a face $\mathcal{Q}_{t,v}^\mathcal{F}$. Interior points $\mu_3 \in \mathcal{M}_{t,v}^\circ$ are exactly those reproducible by positive distributions with respect to ν .

$$\Lambda^{\delta, \text{MAX}, \nu} = \text{conv} \left(\epsilon_{x_{[d]}}[X_{[d]}] : \nu[X_{[d]} = x_{[d]}] = 1 \right).$$

This follows from the fact, that the support of any by ν representable distribution is contained in the support of ν . Further, each representable distribution is contained in the convex hull of the one-hot encoded support elements, since any distribution is normalized.

The polytope of mean parameters is a linear transform of the elements in $\Lambda^{\delta, \text{MAX}, \nu}$, since the contraction with σ^t is linear. It follows that

$$\begin{aligned} \mathcal{M}_{t,v} &= \text{conv} \left(\langle \sigma^t[X_{[d]}, L], \epsilon_{x_{[d]}}[X_{[d]}] \rangle_{[L]} : \nu[X_{[d]} = x_{[d]}] = 1 \right) \\ &= \text{conv} \left(\sigma^t[X_{[d]} = x_{[d]}, L] : \nu[X_{[d]} = x_{[d]}] = 1 \right). \square \end{aligned}$$

We thus understand the map

$$\sigma^t : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathcal{M}_{t,v} \quad , \quad x \rightarrow \sigma^t[X_{[d]} = x_{[d]}, L]$$

as an encoding of states with respect to a statistic t . When the statistic is the universal statistic $t = \delta$, this statistic encoding coincides with the one-hot encoding and the mean polytope is the simplex of dimension $\langle \nu \rangle_{[\emptyset]} - 1$

$$\mathcal{M}_{\delta,v} = \text{conv} \left(\epsilon_{x_{[d]}}[X_{[d]}] : \nu[X_{[d]} = x_{[d]}] = 1 \right).$$

A generic mean polytope $\mathcal{M}_{t,v}$ is then the linear transform of the simplex by the contraction with $\sigma^t[X_{[d]}, L]$, where L is left open.

1.1.2 Representation as Intersecting Half-Spaces

For any vector $a[L] \in \mathbb{R}^P$ and a scalar $b \in \mathbb{R}$, we call the set

$$\{\mu[L] : \langle \mu[L], a[L] \rangle_{[\emptyset]} \leq b\} \subset \mathbb{R}^P$$

a half-space of $\mathbb{R}^P$. Bounded intersections of finitely many half-spaces are called $\mathcal{H}$ -polytopes [?]. We state next, that the polytope $\mathcal{M}_{t,\nu}$ of mean parameters is a $\mathcal{H}$ -polytopes.

Theorem 1.2 *The set $\mathcal{M}_{t,\nu}$ is for any statistic t and base measure ν a $\mathcal{H}$ -polytope, i.e. there exists a finite collection*

$$((a_i[L], b_i) : i \in [n])$$

where $a_i[L]$ a vector and $b_i \in \mathbb{R}$ for all $i \in [n]$ such that

$$\mathcal{M}_{t,\nu} = \{\mu[L] : \forall_{i \in [n]} \langle \mu[L], a_i[L] \rangle_{[\emptyset]} \leq b_i\}.$$

Proof. By Thm. 1.1, the set $\mathcal{M}_{t,\nu}$ is the convex hull of a finite set of vectors and is therefore a $\mathcal{V}$ -polytope. We therefore apply the main theorem for polytopes [?], which states the equivalence of $\mathcal{V}$ -polytopes and $\mathcal{H}$ -polytope, for which a proof can be found as Theorem 1.1 in [?]. Therefore, $\mathcal{M}_{t,\nu}$ is also a $\mathcal{H}$ -polytope and has is thus the intersection of finitely many half-spaces. $\square$

The determination of the half-space parametrizing $((a_i[L], b_i) : i \in [n])$ is, however, in general difficult and the main reason for the intractability of probabilistic inference (see e.g. [?]).

1.1.3 Characterization of the Interior

The interior of the mean polytope consists of the mean parameters to positive distributions as we show next.

Theorem 1.3 *For any minimal statistics t with respect to a boolean base measure ν (see Def. ??) and a with respect to ν positive distribution $\mathbb{P}[X_{[d]}]$ we have*

$$\langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L] \rangle_{[L]} \in (\mathcal{M}_{t,\nu})^\circ.$$

Proof. Since by assumption the statistics is minimal, the convex set $\mathcal{M}_{t,\nu}$ is full dimensional (see e.g. Appendix B in [?]). We thus use a well-known property for full-dimensional convex sets (see [?, ?]), that $\mu \in \mathcal{M}_{t,\nu}^\circ$ if for any non-vanishing vector $V[L]$ there is a $\tilde{\mu}[L]$ with

$$\langle V[L], \mu[L] \rangle_{[\emptyset]} < \langle V[L], \tilde{\mu}[L] \rangle_{[\emptyset]}.$$

It thus suffices to show for an arbitrary non-vanishing vector $V[L]$ the existence of a distribution $\tilde{\mathbb{P}}$, such that

$$\langle V[L], \mu[L] \rangle_{[\emptyset]} < \langle V[L], \sigma^t[X_{[d]}, L], \tilde{\mathbb{P}}[X_{[d]}] \rangle_{[\emptyset]}.$$

We define for $\epsilon \in \mathbb{R}$

$$\mathbb{P}^\epsilon[X_{[d]}] = \left\langle \mathbb{P}[X_{[d]}], \exp \left[\epsilon \cdot \langle \sigma^t[X_{[d]}, L], V[L] \rangle_{[X_{[d]}]} \right] \right\rangle_{[X_{[d]}|\emptyset]}$$

The derivation of this map at $\epsilon = 0$ is

$$\frac{\partial}{\partial \epsilon} \mathbb{P}^\epsilon[X_{[d]}] |_{\epsilon=0} = \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L], V[L] \rangle_{[X_{[d]}]} - \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L], V[L] \rangle_{[\emptyset]} \cdot \mathbb{P}[X_{[d]}]$$

and thus

$$\begin{aligned} \frac{\partial}{\partial \epsilon} \langle \mathbb{P}^\epsilon[X_{[d]}], \sigma^t[X_{[d]}, L], V[L] \rangle_{[\emptyset]} |_{\epsilon=0} &= \left\langle \mathbb{P}[X_{[d]}], (\langle \sigma^t[X_{[d]}, L], V[L] \rangle_{[1]})^2 \right\rangle_{[X_{[d]}]} \\ &\quad - \left(\langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L], V[L] \rangle_{[X_{[d]}]} \right)^2. \end{aligned}$$

We can interpret this quantity as the variance of the random variable

$$\langle \sigma^t [X_{[d]}, L], V[L] \rangle_{[X_{[d]}=x_{[d]}]} ,$$

where $x_{[d]}$ is drawn from $\mathbb{P} [X_{[d]}]$. The variance is greater than zero, if this random variable is not constant. But from the minimality of t with respect to ν it follows, that this variable is not constant and we therefore have

$$0 < \frac{\partial}{\partial \epsilon} \langle \mathbb{P}^\epsilon [X_{[d]}], \sigma^t [X_{[d]}, L], V[L] \rangle_{[\emptyset]} |_{\epsilon=0} .$$

Thus, there is a $\epsilon > 0$ with

$$\langle V[L], \mu [L] \rangle_{[\emptyset]} < \langle V[L], \sigma^t [X_{[d]}, L], \mathbb{P}^\epsilon [X_{[d]}] \rangle_{[\emptyset]} . \square$$

While Thm. 1.3 only states that the mean parameter of each positive distribution is in the interior, we will construct for each interior point a positive distribution in Chapter ?? by a member of the corresponding exponential family.

1.1.4 Characterization of the Boundary by Faces

Let us now continue with the investigation of the faces of the mean parameter polytope.

Definition 1.2 Given a mean parameter polytope $\mathcal{M}_{t,\nu}$ in the half space representation of Thm. 1.2, and any subset $\mathcal{I} \subset [n]$ we say that the set

$$\mathcal{Q}_{t,\nu}^{\mathcal{I}} = \{ \mu [L] \in \mathcal{M}_{t,\nu} : \forall_{i \in \mathcal{I}} \langle \mu [L], a_i [L] \rangle_{[\emptyset]} = b_i \}$$

is the face to the constraints $\mathcal{I}$.

While all inequalities in a half-space representation are satisfied for any element of the polytope, we defined faces by the additional sharp satisfaction of a subset of the half-space inequalities. In this way, the faces build the boundary of $\mathcal{M}_{t,\nu}$. This can be easily verified, since for any vector $\mu [L] \in \mathcal{M}_{t,\nu}$, for which no halfspace inequalities hold sharply, also a neighborhood satisfies the halfspace inequalities. If any halfspace inequality holds sharply, in the other case, the vector is a member of the corresponding face.

If t is not minimal with respect to ν , we find a non-vanishing vector $V[L]$ and a scalar $\lambda \in \mathbb{R}$ such that

$$\langle \sigma^t [X_{[d]}, L], V[L], \nu [X_{[d]}] \rangle_{[X_{[d]}]} = \lambda \cdot \nu [X_{[d]}] .$$

This implies, that any probability distribution $\mathbb{P} [X_{[d]}]$ representable with ν satisfies

$$\langle \mathbb{P} [X_{[d]}], \sigma^t [X_{[d]}, L], V[L], \nu [X_{[d]}] \rangle_{[\emptyset]} = \lambda \cdot \langle \mathbb{P} [X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]} = \lambda .$$

Any $\mu [L] \in \mathcal{M}_{t,\nu}$ then satisfies

$$\langle \mu [L], V[L] \rangle_{[\emptyset]} = \lambda .$$

Thus, the polytope $\mathcal{M}_{t,\nu}$ is contained in an affine linear subspace and has vanishing interior. We can further understand this equation as two half-space inequalities

$$\langle \mu [L], V[L] \rangle_{[\emptyset]} \leq \lambda \quad \text{and} \quad \langle \mu [L], V[L] \rangle_{[\emptyset]} \geq \lambda ,$$

which can be integrated into any half-space representation. We conclude, that in the case of non-minimal statistics, the whole polytope $\mathcal{M}_{t,\nu}$ is a face itself, since it satisfies these half-space inequalities sharply.

Lemma 1.1 For each face $\mathcal{Q}_{t,\nu}^{\mathcal{I}}$ we have

$$\mathcal{Q}_{t,\nu}^{\mathcal{I}} = \text{conv} \left(\sigma^t [X_{[d]} = x_{[d]}, L] : x_{[d]} \in (\sigma^t)^{-1}(\mathcal{Q}_{t,\nu}^{\mathcal{I}}), \nu [X_{[d]} = x_{[d]}] = 1 \right) .$$

Proof. This holds, since each face is the convex hull of the contained vertices (see Proposition 2.2 and 2.3 in [?]). Since the vertices are contained in the image of the statistic encoding σ^t , the vertices contained in $Q_{t,v}^{\mathcal{F}}$ are contained in the set

$$\sigma^t [X_{[d]} = x_{[d]}, L] : x_{[d]} \in (\sigma^t)^{-1}(Q_{t,v}^{\mathcal{F}}) . \square$$

Lem. 1.1 implies in particular, that faces are mean parameter polytopes with respect to refined base measures. For reference in later chapters, we define these refined base measures next as face measures.

Definition 1.3 The face measure to the face $Q_{t,v}^{\mathcal{F}}$ of $\mathcal{M}_{t,v}$ is the boolean tensor $\nu^{t,\mathcal{F}} [X_{[d]}]$ with coordinates to $x_{[d]} \in \times_{k \in [d]} [m_k]$ by

$$\nu^{t,\mathcal{F}} [X_{[d]} = x_{[d]}] = \begin{cases} 1 & \text{if } \gamma^{x_{[d]}} \in Q_{t,v}^{\mathcal{F}} \\ 0 & \text{else} \end{cases} .$$

We now specify the mean parameter polytope to any face using the face measure as a refinement of the base measure.

Theorem 1.4 For any face $Q_{t,v}^{\mathcal{F}}$ of $\mathcal{M}_{t,v}$, we have with the refined base measure

$$\tilde{\nu} [X_{[d]}] = \langle \nu [X_{[d]}], \nu^{t,\mathcal{F}} [X_{[d]}] \rangle_{[X_{[d]}]}$$

that

$$Q_{t,v}^{\mathcal{F}} = \mathcal{M}_{t,\tilde{\nu}} .$$

Proof. We notice that for any $x_{[d]} \in \times_{k \in [d]} [m_k]$, $x_{[d]} \in (\sigma^t)^{-1}(Q_{t,v}^{\mathcal{F}})$ is equal to $\nu^{t,\mathcal{F}} [X_{[d]} = x_{[d]}] = 1$ and thus

$$\{x_{[d]} : x_{[d]} \in (\sigma^t)^{-1}(Q_{t,v}^{\mathcal{F}}), \nu [X_{[d]} = x_{[d]}] = 1\} = \{x_{[d]} : \tilde{\nu} [X_{[d]} = x_{[d]}] = 1\} .$$

In combination with Lem. 1.1 we then get

$$\begin{aligned} Q_{t,v}^{\mathcal{F}} &= \text{conv} \left(\sigma^t [X_{[d]} = x_{[d]}, L] : x_{[d]} \in (\sigma^t)^{-1}(Q_{t,v}^{\mathcal{F}}), \nu [X_{[d]} = x_{[d]}] = 1 \right) \\ &= \text{conv} \left(\sigma^t [X_{[d]} = x_{[d]}, L] : x_{[d]} : \tilde{\nu} [X_{[d]} = x_{[d]}] = 1 \right) \\ &= \mathcal{M}_{t,\tilde{\nu}} . \square \end{aligned}$$

Positivity of a distribution with respect to face measures is an equivalent condition for the mean parameter of a distribution to be on a face, as we show next.

Theorem 1.5 If and only if for a distribution $\mathbb{P} [X_{[d]}]$ and a face $\mathcal{F}$ we have

$$\langle \mathbb{P} [X_{[d]}], \sigma^t [X_{[d]}, L] \rangle_{[L]} \in Q_{t,v}^{\mathcal{F}} ,$$

then $\mathbb{P} [X_{[d]}]$ is representable with respect to the base measure

$$\langle \nu [X_{[d]}], \nu^{t,\mathcal{F}} [X_{[d]}] \rangle_{[X_{[d]}]} .$$

Proof. We have

$$\mu [L] = \sum_{x_{[d]}} \mathbb{P} [X_{[d]} = x_{[d]}] \cdot \gamma^t [X_{[d]} = x_{[d]}, L] .$$

Now, the $x_{[d]}$ with $\nu^{t,\mathcal{F}} [X_{[d]} = x_{[d]}] = 1$ are exactly those, for which the conditions $\mathcal{F}$ hold straight. If and only if for a $x_{[d]}$ with $\nu^{t,\mathcal{F}} [X_{[d]} = x_{[d]}] = 0$ we have $\mathbb{P} [X_{[d]} = x_{[d]}] > 0$, one of the conditions $\mathcal{F}$ would not hold straight. Thus, if and only if $\mathbb{P} [X_{[d]}]$ is representable with respect to $\nu^{t,\mathcal{F}} [X_{[d]}]$, we have $\mu [L] \in Q_{t,v}^{\mathcal{F}}$. $\square$

For members of exponential families, we can make a stronger statement than Thm. 1.5. If for any $\mathbb{P}[X_{[d]}] \in \Gamma^{t,\nu}$ and a face $\mathcal{F}$ we have $\langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L] \rangle_{[L]} \in (Q_{t,\nu}^{\mathcal{F}})^\circ$ then $\mathbb{P}[X_{[d]}]$ is positive with respect to the base measure

$$\langle \nu[X_{[d]}], \nu^{t,\mathcal{F}}[X_{[d]}] \rangle_{[X_{[d]}]} .$$

Let us now investigate tensor network representations of face measures, based on the basis encoding β^t of a statistic. Vertices of $\mathcal{M}_{t,\nu}$ are faces with single elements, that is $\{\mu[L]\}$. By Lem. 1.1 there must be μ must lie in the image of σ^t , since otherwise $\mathcal{M}_{t,\nu}$ would be empty. The vertex measure is then

$$\nu^{t,\mathcal{F}}[X_{[d]}] = \langle \beta^t[Y_{[p]}, X_{[d]}], \epsilon_\mu[Y_{[p]}] \rangle_{[X_{[d]}]}$$

Here we use that each $\mu \in Q_{t,\nu}^{\mathcal{F}} \cap \text{im}(\sigma^t)$ has integer-valued coordinates and denote

$$\epsilon_\mu[Y_{[p]}] = \bigotimes_{\ell \in [p]} \epsilon_{\mu[L=\ell]}[Y_\ell] .$$

Theorem 1.6 (Face measure representation) *For any face $Q_{t,\nu}^{\mathcal{F}}$ of $\mathcal{M}$ we have*

$$\nu^{t,\mathcal{F}}[X_{[d]}] = \langle \beta^t[Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}}[Y_{[p]}] \rangle_{[X_{[d]}]}$$

where

$$\kappa^{\mathcal{F}}[Y_{[p]}] = \sum_{\mu \in Q_{t,\nu}^{\mathcal{F}} \cap \text{im}(\sigma^t)} \epsilon_\mu[Y_{[p]}] .$$

Proof. For any $\mu \in Q_{t,\nu}^{\mathcal{F}} \cap \text{im}(\sigma^t)$ the tensor

$$\tau^\mu[X_{[d]}] = \langle \beta^t[Y_{[p]}, X_{[d]}], \epsilon_\mu[Y_{[p]}] \rangle_{[X_{[d]}]}$$

is the indicator of the preimage of μ under σ^t . Since preimages the elements in $Q_{t,\nu}^{\mathcal{F}} \cap \text{im}(\sigma^t)$ are disjoint, the support of $\tau^\mu[X_{[d]}]$ is disjoint and their sum

$$\sum_{\mu \in Q_{t,\nu}^{\mathcal{F}} \cap \text{im}(\sigma^t)} \tau^\mu[X_{[d]}]$$

is the indicator of the preimage of $Q_{t,\nu}^{\mathcal{F}}$ under σ^t , which is the face measure $\nu^{t,\mathcal{F}}[X_{[d]}]$. Exploiting linearity of contraction we have

$$\begin{aligned} \nu^{t,\mathcal{F}}[X_{[d]}] &= \sum_{\mu \in Q_{t,\nu}^{\mathcal{F}} \cap \text{im}(\sigma^t)} \tau^\mu[X_{[d]}] \\ &= \left\langle \beta^t[Y_{[p]}, X_{[d]}], \sum_{\mu \in Q_{t,\nu}^{\mathcal{F}} \cap \text{im}(\sigma^t)} \epsilon_\mu[Y_{[p]}] \right\rangle_{[X_{[d]}]} \\ &= \langle \beta^t[Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}}[Y_{[p]}] \rangle_{[X_{[d]}]} . \square \end{aligned}$$

Let us now investigate, which normalized face measures can be computed using t and a hypergraph $\mathcal{G}$.

Example 1.1 (Vertices) Vertices $Q_{t,\nu}^{\mathcal{F}}$ are proper faces of affine dimension 0, that is they consist in single vectors. Since all vertices are in the image $\sigma^t(X)$, there exists an index tuple $x_{[d]} \in X$ such that $\nu[X_{[d]} = x_{[d]}] = 1$ and

$$Q_{t,\nu}^{\mathcal{F}} = \{\sigma^t[X_{[d]} = x_{[d]}, L]\} .$$

Then $\kappa^{\mathcal{F}}[Y_{[p]}]$ is the one-hot encoding of the by an interpretation map I assigned index to $\sigma^t[X_{[d]} = x_{[d]}, L]$, that is

$$\kappa^{\mathcal{F}} [Y_{[p]}] = \epsilon_{I^{-1}(\sigma^t[X_{[d]=x_{[d]}, L])} [Y_{[p]}] .$$

In particular, the activation core is elementary and the face measure to any vertex is in $\Lambda^{t, \text{EL}, \nu}$.

Extending Example 1.1, we can provide a coarse estimation of the hypergraph $\mathcal{G}$ required to decompose $\kappa^{\mathcal{F}}$ for generic faces $Q_{t, \nu}^{\mathcal{F}}$. We notice that $\kappa^{\mathcal{F}} [Y_{[p]}]$ in Thm. 1.6 is a sparse tensor with basis CP rank $|Q_{t, \nu}^{\mathcal{F}} \cap \text{im}(\sigma^t)|$ (see Chapter ??).

$$\ell_0(\kappa^{\mathcal{F}} [Y_{[p]}]) = |\sigma^t(\nu^{t, \mathcal{F}})|$$

where $|\sigma^t(\nu^{t, \mathcal{F}})|$ is the number of different statistic encoding vectors to the support of the face measure $\nu^{t, \mathcal{F}}$. Using the formalism of sparse CP decompositions Chapter ??, this characterizes the basis CP rank of $\kappa^{\mathcal{F}} [Y_{[p]}]$. However, the basis CP rank is only an upper bound to generic CP rank, which can be loose. By Example 1.2 we provide with the maximal face an example, where the basis CP rank is given by the tensor space dimension, whereas the generic CP rank is one and the normalized face measure is thus still in $\Lambda^{t, \text{EL}}$.

Example 1.2 (Maximal face) The maximal face $Q_{t, \nu}^{\emptyset} = \mathcal{M}_{t, \nu}$ coincides with the mean polytope itself. In this case the corresponding activation tensor to the face measure is trivial, that is

$$\kappa^{\emptyset} [Y_{[p]}] = \mathbb{I} [Y_{[p]}] .$$

$\kappa^{\emptyset}$ is elementary and the normalized face measure $\nu^{t, \emptyset}$ to the maximal face is in $\Lambda^{t, \text{EL}}$.

1.1.5 Partition into Effective Interiors of Faces

Let us now introduce effective interiors, which enables us to find disjoint partitions of the mean polytope.

Definition 1.4 (Effective Interior) Let $\mathcal{U} \subset \mathbb{R}^p$ be an arbitrary set and $\mathcal{L}$ the minimal affine subspace of $\mathbb{R}^p$ containing $\mathcal{U}$. Then the effective interior, denoted $(\mathcal{U})^\circ$ is the interior of $\mathcal{U}$ in the space $\mathcal{L}$.

Lemma 1.2 Any polytope is a disjoint union of the effective interiors of its faces, that is

$$\mathcal{M}_{t, \nu} = \bigcup_{\mathcal{F} \subset [n]} (Q_{t, \nu}^{\mathcal{F}})^\circ .$$

Proof. For any $\mu \in \mathcal{M}_{t, \nu}$ we find a face such that $\mu \in Q_{t, \nu}^{\mathcal{F}}$. If $\mu \notin (Q_{t, \nu}^{\mathcal{F}})^\circ$, then there is a face $Q_{t, \nu}^{\tilde{\mathcal{F}}} \subset Q_{t, \nu}^{\mathcal{F}}$ of smaller affine dimension such that $\mu \in Q_{t, \nu}^{\tilde{\mathcal{F}}}$. When continuing this process we reach a face such that $\mu \in (Q_{t, \nu}^{\tilde{\mathcal{F}}})^\circ$, since the faces with affine dimension 0 are vertices and they coincide with their effective interior because they contain a single vector. $\square$

In this way, we find to each $\mu \in \mathcal{M}_{t, \nu}$ a unique exponential family with statistics t and base measure by a face measure, such that μ is reproduced by an element of that exponential family. We will show in Chapter ??, that these reproducing distributions maximize the entropy among any other reproducing distribution.

1.1.6 Centers

We provide further intuition on the position of the mean parameter, when interpreting ν and t as soft and hard constraint mechanisms on a distribution. To this end, let $\mathcal{X}$ be an arbitrary finite state set, where we so far had the set $\times_{k \in [d]} [m_k]$ for factored representations.

Definition 1.5 We call the mean parameter of a normalized base measure, that is the vector

$$\mu_{t,0,\nu} [L] = \left\langle \langle \nu \rangle_{[X_{[d]}|\emptyset]}, \sigma^t [X_{[d]}, L] \right\rangle_{[L]}.$$

the center of $\mathcal{M}_{t,\nu}$ (see Figure 1.2).

We have

$$\mu_{t,0,\nu} [L] = \frac{1}{\langle \nu \rangle_{[\emptyset]}} \sum_{x_{[d]} : \nu[X_{[d]}=x_{[d]}]=1} \sigma^t [X_{[d]}=x_{[d]}, L]$$

and therefore understand $\mu_{t,0,\nu}$ as the weighted center of $\mathcal{M}_{t,\nu}$, where weights are allocated on the image of the statistics encoding by the cardinality of their pre-images. These concepts are sketched in Figure 1.2 in blue.

The choice of a base measure ν can be regarded as the hard structure of a distribution. Given ν , we further implement soft structure in form of distributions in exponential families $\Gamma^{t,\nu}$. While the normalized base measure ν reproduces only the weighted center of $\mathcal{M}_{t,\nu}$, canonical parameters can be chosen to alter to any interior point in $\mathcal{M}_{t,\nu}$. This alternations are sketched in Figure 1.2 in red.

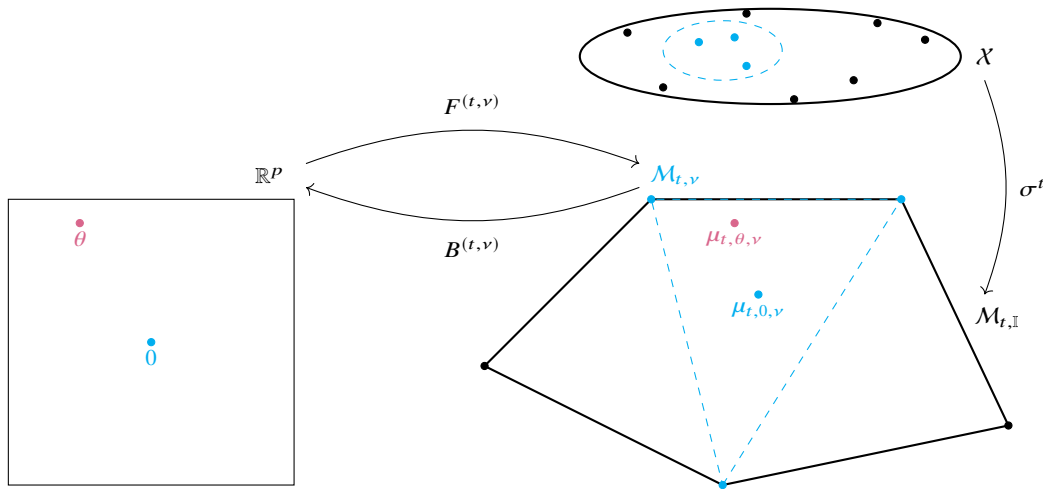


Fig. 1.2 Sketch of hard constraints (blue) and soft constraints (red) determining the position of the mean parameter in $\mu_{t,\mathbb{I}}$. In the set of states $\mathcal{X}$, the support of a base measure ν is marked blue, and the base measure is the corresponding subset encoding. The normalized ν has the mean parameter $\mu_{t,0,\nu}$, and the mean polytope $\mathcal{M}_{t,\nu}$ is the convex hull of the statistics encoding σ^t restricted on the support set of ν . Given a base measure ν , the corresponding forward and backward maps implement soft constraints depending on canonical parameters θ . The forward mapping maps the zero canonical parameter to $\mu_{t,0,\nu}$, correspond with the normalized base measure. Generic parameters θ (red) are mapped onto the interior $\mathcal{M}_{t,\nu}^\circ$.

1.2 Mean as a Statistic

We in this section take the perspective of estimating a maximum entropy distribution given observed data.

- μ_D is an unbiased estimator of μ , i.e. $\mathbb{E} [\mu_D [L]] = \mu$
- μ_D is a consistent estimator of μ , i.e. $\mu_D [L] \rightarrow \mu$ by the law of large numbers coordinatewise almost everywhere.
- μ_D is the minimal variance unbiased estimator of μ .

The mean parameter μ_D given a dataset can be understood as a statistic of the dataset. We here show that for the family of maximum entropy distributions this statistic is a minimal sufficient statistic.

The family of maximum entropy distributions is the set

$$\{\mathbb{P}^\mu [X_{[d]}] : \mu \in \mathcal{M}_{t,v}\}$$

which has been characterized above by a union of exponential families with respect to face measures.

Taking a frequentist perspective we now understand datasets by random variables $X_{[d] \times [m]}$, where for $j \in [m]$ the variables $X_{[d],j}$ are drawn i.i.d. from a maximum entropy distribution. The mean statistic is then a tensor

$$\mu_D [X_{[d] \times [m]}, L]$$

with coordinates

$$\mu_D [X_{[d] \times [m]} = x_{[d] \times [m]}, L] = \frac{1}{m} \sum_{j \in [m]} \sigma^t [X_{[d],j} = x_{[d],j}, L] .$$

Theorem 1.7 *The mean statistic is a sufficient statistic for the family of maximum entropy distributions (t, v) . If the base measure has maximal support, the mean statistic is in addition minimal.*

Proof. It suffices to show that the likelihood is a function of μ_D . Let us choose a face $\mathcal{F}$ of $\mathcal{M}_{t,v}$, then the likelihood is different from 0 if and only if the empirical distribution is representable with respect to the face measure. This is the case if and only if μ_D is on the face. In case that μ_D is on the face, then the likelihood of any distribution on that face exponential family is

$$\exp \left[m \cdot \left(\langle \mu_D [X_{[d] \times [m]}, L], \theta [L] \rangle_{[\emptyset]} - A^{(t,v)}(\theta) \right) \right]$$

We have thus shown that the likelihood is always a function of μ_D .

The minimality claim can be shown by the constant quotient criterion (see Thm. 6.2.13 in [?]).

□

Chapter 2

Variational Inference

Abstract We present here variational inference methods to reason on Computation-Activation Networks.
Topics:

- I.3.4 (Exponential family) Variational inference
- I.3.7 Mean field methods

2.1 Foundational inference problems

We have two parametrizations of maximum entropy distributions:

- **Mean parametrization:** The mean parameter $\mu [L] \in \mathcal{M}_{t,\nu}$
- **Canonical parametrization:** The face $\mathcal{F} \in \mathcal{Q}_{t,\nu}^{\mathcal{F}}$ together with the weight vector $\theta [L]$

The fundamental inference problem are finding one parametrization given the other.

- **Forward mapping:** Given the canonical parametrization $(\mathcal{F}, \theta [L])$, find the mean parameter $\mu [L]$.
- **Backward mapping:** Given the mean parameter $\mu [L]$, find the face $\mathcal{F}$ such that $\mu [L]$ is in the relative interior of the face, and the corresponding canonical parameter $\theta [L]$.

2.1.1 Forward mapping

The forward mapping is the contraction of the distribution with the selection encoding of the statistic, that is the map

$$F^{(t,\nu)}(t) : \mathcal{P}_t \rightarrow \mathcal{M}_t$$

with

$$F^t(\mathcal{F}, \theta) = \frac{\langle \alpha^\theta [Y_{[p]}], \kappa^{\mathcal{F}} [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}], \sigma^t [X_{[d]}, L] \rangle_{[L]}}{\langle \alpha^\theta [Y_{[p]}], \kappa^{\mathcal{F}} [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \rangle_{[\emptyset]}}.$$

2.1.2 Backward mapping

Determining the face such that $\mu [L]$ in the relative interior, and the corresponding canonical parameter.

2.1.2.1 Cross entropy minimization

The cross entropy between a distribution $\mathbb{P}^* [X_{[d]}]$ and a maximum entropy distribution $\mathbb{P}^{\mathcal{F}, \theta} [X_{[d]}]$ is given by

$$\begin{aligned} \mathbb{H} [\mathbb{P}^* [X_{[d]}], \mathbb{P}^{\mathcal{F}, \theta} [X_{[d]}]] &= - \langle \mathbb{P}^* [X_{[d]}], \beta^t [Y_{[p]}, X_{[d]}], \ln [\alpha^\theta [Y_{[p]}]] \rangle_{[\emptyset]} \\ &\quad - \langle \mathbb{P}^* [X_{[d]}], \beta^t [Y_{[p]}, X_{[d]}], \ln [\kappa^{\mathcal{F}} [Y_{[p]}]] \rangle_{[\emptyset]} \\ &\quad + \ln \left[\langle \alpha^\theta [Y_{[p]}], \kappa^{\mathcal{F}} [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \rangle_{[\emptyset]} \right] \end{aligned}$$

The first term can be simplified as

$$\begin{aligned} - \langle \mathbb{P}^* [X_{[d]}], \beta^t [Y_{[p]}, X_{[d]}], \ln [\alpha^\theta [Y_{[p]}]] \rangle_{[\emptyset]} &= - \sum_{\ell \in [p]} \theta [L = \ell] \cdot \langle \mathbb{P}^* [X_{[d]}], t_\ell [X_{[d]}] \rangle_{[\emptyset]} \\ &= - \langle \mu^* [L], \theta [L] \rangle_{[\emptyset]} \end{aligned}$$

To investigate the second term, we notice that

$$\langle \beta^t [Y_{[p]}, X_{[d]}], \ln [\kappa^{\mathcal{F}} [Y_{[p]}]] \rangle_{[X_{[d]}=x_{[d]}]} = \begin{cases} 0 & \text{if } \gamma^{x_{[d]}} \in \mathcal{F} \\ -\infty & \text{else} \end{cases}$$

Therefore the cross entropy is finite, if and only if $\mathbb{P}^* [X_{[d]}]$ is supported at the face measure. This is only the case if $\mu^* [L] \in \mathcal{F}$. If the cross entropy is finite, the second term vanishes.

Among the faces with $\mu^* [L] \in \mathcal{F}$ the third term is minimal for each $\theta \in \mathbb{R}^p$.

$$\begin{aligned} &\operatorname{argmin}_{\mathcal{F} \in L(\mathcal{M}_{t,v})} \operatorname{argmin}_{\theta \in \mathbb{R}^p} \mathbb{H} [\mathbb{P} [X_{[d]}], \mathbb{P}^{\mathcal{F}, \theta} [X_{[d]}]] \\ &= \operatorname{argmin}_{\theta \in \mathbb{R}^p} - \langle \mu^* [L], \theta [L] \rangle_{[\emptyset]} + \ln \left[\langle \alpha^\theta [Y_{[p]}], \kappa^{\mathcal{F}^{\mu^*}} [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \rangle_{[\emptyset]} \right] \end{aligned}$$

2.1.2.2 Definition of backward mapping

2.2 Face estimation

Generic situation: Find the face in $L(\mathcal{M}_t)$ which contains μ^* in its relative interior.

Theorem 2.1 *Let $\mathbb{P}^* [X_{[d]}]$ be a distribution with mean parameter $\mu^* [L]$. Then the face $\mathcal{F}(\mu^*)$ is the minimal face in the lattice $L(\mathcal{M}_t)$ such that*

$$\langle \mathbb{P}^* [X_{[d]}], \beta^t [Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}} [Y_{[p]}] \rangle_{[\emptyset]} = 1.$$

Proof. • $\mu^* [L] \in \mathcal{F}$ is equivalent to $\langle \mathbb{P}^* [X_{[d]}], \beta^t [Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}} [Y_{[p]}] \rangle_{[\emptyset]} = 1$.

• If $\mu^* [L]$ is not in the relative interior of $\mathcal{F}$, then it is contained in a smaller face.

□

2.2.1 Cube-like faces

Cube-faces are parametrized by $\mathcal{U}$ and a boolean tuple $y_{\mathcal{U}}$. For cube-like faces, it is enough to decide entailment and contradiction of each feature.

Theorem 2.2 *If the mean parameter $\mu^* [L]$ to a distribution $\mathbb{P}^* [X_{[d]}]$ is in the relative interior of a cube-like face, then the face $\mathcal{F}(\mu^*)$*

$$\mathcal{U} = \{\ell : \ell \in [p], \langle \mathbb{P}^* [X_{[d]}], t_\ell [X_{[d]}] \rangle_{[\emptyset]} \in \{0, 1\}\}$$

and

$$y_\ell = \langle \mathbb{P}^* [X_{[d]}], t_\ell [X_{[d]}] \rangle_{[\emptyset]} .$$

2.2.2 Face refinement

We present Algorithm 1 to search the $\mathcal{F}(\mu^*)$ in the face lattice. In each iteration of the while loop it either finds a facet of the current with $\mu^*[L] \in \tilde{\mathcal{F}}$ or it terminates. Since the face lattice has a finite depth n , it always terminates after at most n iterations of the while loop.

Algorithm 1 Face refinement

Require: Distribution $\mathbb{P}^* [X_{[d]}]$, statistic t

Ensure: Face $\mathcal{F}(\mu^*[L])$

$\mathcal{F} \leftarrow \mathcal{M}_{t, \nu}$

while True **do**

 refined $\leftarrow$ False

for all facets $\tilde{\mathcal{F}}$ of $\mathcal{F}$ **do**

if not refined **then**

 Compute

$$\lambda = \langle \mathbb{P}^* [X_{[d]}], \nu^{\tilde{\mathcal{F}}} [X_{[d]}] \rangle_{[\emptyset]}$$

if $\lambda = 1$ **then**

 refined $\leftarrow$ True

$\mathcal{F} \leftarrow \tilde{\mathcal{F}}$

end if

end if

end for

if not refined **then**

return $\mathcal{F}$

end if

end while

2.3 Parameter estimation

We now present iterative algorithms to train the weight vector, once the face to a mean parameter has been determined. They are coordinate descent methods on the cross-entropy loss function.

2.3.1 Moment matching

We differentiate the cross-entropy with respect to the weight vector

$$\frac{\partial \mathbb{H} [\mathbb{P}^*, \mathbb{P}^{\mathcal{F}, \theta}]}{\partial \theta [L = \ell]} = \mu [L = \ell] - \mu^* [L = \ell] .$$

The derivative vanishes when both mean parameters coincide, which is the moment matching condition.

2.3.2 Issues with erroneous faces

If applied to the wrong face, they fail:

- If $\mu [L]$ not on the face, then the cross-entropy is infinite.
- If $\mu [L]$ on the face, but not in the relative interior, then the norm of the weight vector diverges.